INCH-POUND

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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

STRAINER, SIMPLEX, CONTINUOUS SELF-CLEANING

The General Service Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE

1.1 Scope. This commercial item description covers continuous self-cleaning simplex strainers.

1.2 Intended use. The strainers specified are intended for installation on the discharge side of the pump in the fuel/lubricating oil piping systems.

2. CLASSIFICATION

2.1 Classification. Units should be of the following types and classes as specified (see 7.1):

Type I - 150 pound per square inch gage (psig) pressure rating.
Type II - 300 psig pressure rating.
Type III - 600 psig pressure rating.

Class 1 - 75 Micron rating.
Class 2 - 40 Micron rating.
Class 3 - 25 Micron rating.

3. SALIENT CHARACTERISTICS

3.1 Codes and standards. Strainer shall comply with the applicable requirements of table I.

Beneficial comments recommendations, additions, deletions, clarifications etc. and any other data which may improve this document should be sent to: Commander, SEA 03R42, Naval Sea Systems Command, 2531 Jefferson Davis Hwy, Arlington, VA 22242-5160.

AMSC N/A FSC 4730
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TABLE I. Reference data.

Equipment/Characteristic	Description
Fluid type	Fuel, lubricating oil
Operation pressure (Design)	150, 300, 600 psi
Temperature range	70 to 250°F
Piping connections	
Inlet, outlet, screen compartment and sump clean-out.	150, 300, 600 rating ANSI B16.5 full face flanges
Gage, sump drain and vent	SAE J 1926
Pressure resistance	FCI 78-1
Flow capacity	Para 3.4 per ISA-S75.02
Shock resistance	MIL-S-901 grade A
Vibration resistance	MIL-STD-167-1 type I
Airborne noise	MIL-STD-740-1 grade C
Motion & attitude resistance	MIL-STD-1399 section 301

3.2 Design. Each self-cleaning simplex type strainer shall remove particulate contaminants from fuel or lube oil. The self-cleaning action shall be continuous, and shall have the ability to periodically backwash the strained fluid through the strainer element in segments such that the particulate contaminants are flushed free of the strainer element and down into a large sump. The self-cleaning action of each unit shall be driven by a hydraulic motor which uses strained system fluid as its sole source of power.

3.3 General construction. Each strainer assembly shall consist of a housing, strainer element(s) and self-cleaning mechanism(s). The cross-sectional flow area throughout the unit shall be equal to or greater than that of the piping to which the strainer is connected.

3.3.1 Cover. Cover shall be secured to the housing by nuts and bolts. The cover shall effect a positive seal on a gasket or O-ring recessed in the cover. Removal of the cover shall provide access to all internal components and shall not require unbolting the inlet and outlet strainer piping connections. Cover lifting devices with integral supports shall be required for covers weighing more than 30 pounds. The strainer cover shall contain a device to deflect fluid spray downward in the event of seal failure. The spray deflector shall remain in position at all times while the strainer is pressurized.

3.3.1.1 Spray deflector effectiveness test. One strainer of each type, class, size and specific design shall be subjected to this test. The strainer shall be centered in drip pan 3-inches deep and of a size and shape to allow 4-inches clearance around the periphery. The strainer cover shall be installed with an O-ring which has been notched to cause leakage. The strainer shall be pressurized to maximum design pressure. This test shall be conducted 8 times with the O-ring notch moved clockwise 45 degrees between tests. Failure to pass this test in any leakage position shall require redesign of the deflector and retest.

3.3.1.2 Inter-lock feature. The strainer cover shall contain a locking device to prevent disassembly while the unit is pressurized.

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3.3.2 Strainer element construction. Strainer element shall be screen type construction, consisting of a supporting framework (strength member) with a wire screen as the straining medium. The supporting framework (strength member) shall be plate construction. The strainer element shall be made of corrosion-resistant steel. Wire screen edges and seams shall be folded back upon themselves and welded or brazed to preclude fraying. All seam welds shall be continuous. The strainer element shall be held securely in place when the strainer is assembled.

3.3.2.1 Element support. The element support shall be designed to facilitate easy removal/installation of the straining element and element support as an assembly. The element support must satisfy the flow area requirements of section 3.4.

3.3.2.2 Straining element. Unless otherwise specified (see 7.1), the straining element shall be attached to the element support in such a manner that it can easily be removed from the element support and replaced. The free flow area through the assembly of the strainer element and element support shall be not less than 2 times the cross-sectional area of the strainer discharge connection. The element micron rating shall be specified by 2.1. Unless otherwise specified (see 7.1), the minimum beta ratio shall be 75.

3.3.2.2.1 Straining element compression test. A differential pressure (outside-inside) shall be applied across the support/straining element assembly equal to 100 percent of the specified strainer design pressure. This differential may be accomplished by installing a strippable film on the straining element exterior. Then increasing the pressure to the maximum pressure drop and held for 20 minutes. Alternate methods of obtaining differential pressure will be considered by the command or agency concerned. Supporting or straining elements which distort, stretch or exhibit other weakness shall be rejected.

3.3.2.2.2 Straining element expansion test. A differential pressure (inside-outside) shall be applied across the straining element assembly equal to 100 percent of the specified strainer design pressure. This may by accomplished by installing a strippable film on the straining element interior. Then increasing the pressure to the maximum pressure drop and held for 5 minutes. Alternate methods of obtaining differential pressure will be considered by the command or agency concerned. Straining elements which distort, stretch or exhibit other weaknesses shall be rejected.

3.3.3 Self-cleaning mechanism. The self-cleaning action of each unit shall be operated by a hydraulic motor included as a part of the strainer. Each hydraulic motor shall use strained system fluid as its sole source of power. The fluid consumption of the motor(s) shall not exceed 4 percent of the total flow through the strainer assembly.

3.3.3.1 Backflush structure. The backflush structure shall periodically clean the entire strainer element by backflushing system fluid through the straining element. The structure shall clean the strainer element in segments, so as not to disrupt the system flow through the strainer at any time. All impurities separated shall be deposited within a sludge compartment isolated from the strained liquid.

3.3.4 Sludge compartment. Unless otherwise specified (see 7.1), the sludge compartment shall have a volume of not less than 320 cubic inches for each unit. Sludge compartment shall be equipped with a removable cover for manual cleanout.

3.3.5 Material. Unless otherwise specified (see 7.1), materials shall be as specified in table II.

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TABLE II. Materials for strainers.

Part	Material	Applicable Documents	Remarks
Bodies, covers	CRES	ASTM A 276	S31600
	CRES Plate	ASTM A 167	S30400
Self-cleaning mechanism	CRES Plate	ASTM A 167	S30400
	CRES	ASTM A 276	
Threaded fasteners	Stainless Steel	ASTM F 593	S30400, S31600
		ASTM F 594	
"O" rings	Fluorocarbon	MIL-R-83248/1	
Strainer element	CRES Plate	ASTM A 167	S30400
	CRES Wire	ASTM E 437	S30400, S31600

3.3.6 Envelope dimensions. Unless otherwise specified (see 7.1), dimensions shall be as specified on figure 1.

3.4 Functional performance. Unless otherwise specified (see 7.1), design flow capacity, pressure shall be as follows.

3.4.1 Capacity. The design flow rate for a single element housing shall be 150 gpm for fluids with viscosities less than or equal to 500 Seconds Saybolt Universal (SSU). Additional capacity can be achieved by specifying multiple elements within a single housing (see 7.1).

3.4.1.1 Capacity test. Unless otherwise specified (see 7.1), the test fluid viscosity shall be within ten percent of 500 Seconds Saybolt Universal (SSU). Within this range, test results may be analytically corrected for the deviation of the viscosity from the specified value. The temperature of the fluid must be less than the maximum temperature specified in table I. The strainer assembly shall be installed in a test loop in accordance with ISA-S75.02.

3.4.2 Pressure. Unless otherwise specified (see 7.1), the strainer shall be suitable for operation in the range of 60 psig to design pressure (see 2.1).

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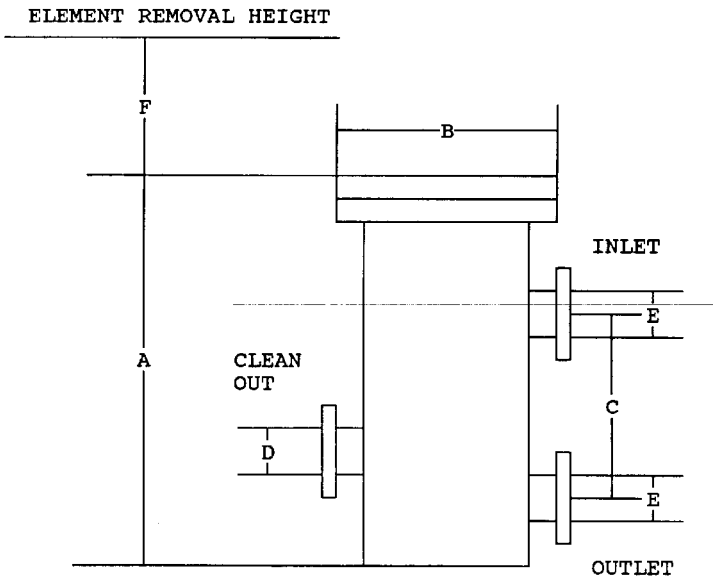

Rated flow (gpm)	75	150	300	450	600	750	900	1050	1200
A	51	51	51	53	54	54	55	55	56
B	10-7/8	10-7/8	27-1/2	27-1/2	32	32	36	36	42
C	30	30	30	30	32	32	36	36	36
D	4	4	5	6	8	8	10	10	10
E	2	3	4	6	6	6	8	8	8
F	25	25	25	25	25	25	25	25	25

FIGURE 1. Envelope dimensions.

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NOTE: Inlet, outlet and cleanout connection dimensions are in nominal pipe size.

All dimensions are in inches.

Flange drilling, straddles center line.

Unless otherwise specified the motor exhaust flange shall be 1-1/2".

Unless otherwise noted all dimensions ± 1/4".

Foundation base shall be designed to accommodate 4 mounting bolts, 1" diameter, 90 degree apart, on a circle of diameter "B".

3.4.3 Self-cleaning rate. The pressure drop through the strainer assembly shall not be greater than 3 psi, unless otherwise specified (see 7.1), during operation. Specifically, the self-cleaning mechanism shall be capable of maintaining the pressure drop below this value when contaminated fluid is being pumped through the strainer at the design flow specified in 3.4.1.

3.4.3.1 Straining element efficiency test. The elements shall be subjected to a strainer element efficiency test using the specified test dust in table III and the test procedure specified in ANSI B93.31.

3.4.3.2 Inclined operation test. The strainer assembly shall be inclined at an angle from the normal equal to the combination of the maximum permanent list and trim as specified in MIL-STD-1399 section 301. The criteria for acceptance shall be as specified in table I.

3.5 Test contaminant. The test dust contaminant shall be of the proposed ISO test dust test grade identified in table III. The dust is available from Powder Technology Inc., (PTI), P.O. Box 1464, Burnsville, MN 55337.

3.6 Workmanship. The strainer and its components shall be free from blow holes, porosity, hard spots, shrinkage defects and cracks. All surfaces shall be smooth and clean. Where dimensions and tolerances affect interchangeability, operation, or performance, they shall be held.

TABLE III. Test dust contaminant.

Micrometres size	Volume % content of test dust larger than	
	Coarse	Medium
5	88 - 89	79 - 82
10	77 - 79	54 - 58
20	61 - 64	29 - 33
40	38 - 41	11 - 13
80	10 - 12	0 - 1
Straining rating (micrometres)	Test dust grade	
10 - 30	Medium	
30 - 80	Coarse	

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4.2 Prohibited material. Zinc or cadmium plating shall not be used on any part of the strainer.

4.3 Mercury exclusion. The strainer shall be free of mercury contamination. During the manufacturing process, tests and examinations (see 7.3), the product to be offered for acceptance shall not come in direct contact with mercury or any of its components, nor with any mercury containing device employing a single boundary of containment.

5. QUALITY ASSURANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

5.2 Additional inspection. Where other specifications form a part of this commercial item description, sampling, examination and tests shall be conducted as required by the pertinent specification, unless otherwise specified in the contract or order (see 7.1).

6. PACKAGING

6.1 Commercial. Commercial packaging shall be in accordance with ASTM D 3951.

6.1.1 Container modification. Shipping containers exceeding 200 pounds gross weight shall be provided with a minimum of two, 3- by 4-inch nominal wood skids laid flat, or a skid- or sill-type base which will support the material and facilitate handling by mechanical handling equipment during shipment, stowage and storage.

7. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

7.1 Acquisition requirements.

(a) Title, number, and date of this commercial item description.
(b) Type and class required (see 1.2).
(c) Issue of DoDISS to be cited in the solicitation, and if required, the specified issue of individual documents referenced (see 7.2.1.1 and 7.2.2).
(d) Type and size of inlet and outlet connections and whether flanged or not flanged (see table I).
(e) Whether gage connections are required at inlet and outlet (see table I).
(f) The required design flow capacity, design pressure, range of fluid temperature and viscosity, and maximum allowable pressure drop (see table I).
(g) Alternate strainer element design and characteristics (see 3.3.2.2).
(h) Alternate or additional design characteristics (see 3.3.4, 3.3.5 and 3.3.6).
(i) Quantity of technical manuals and drawings required (see 7.3).
(j) Whether repair parts are required (see 7.5.1).

7.2 Sources of documents.

7.2.1 Government documents. The following documents of the issue in effect on date of invitation for bids of request for proposal, form a part of the commercial item description to the extent specified.

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7.2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 7.1).

SPECIFICATIONS

DEPARTMENT OF DEFENSE

MIL-S-901 - Shock Tests, H.I. (High Impact): Shipboard Machinery, Equipment and Systems, Requirements For.
MIL-R-83248 - Rubber Fluorocarbon Elastomer, High Performance Fluid, and Compression Set Resistant.
MIL-R-83248/1 - Rubber Fluorocarbon Elastomer, High Temperature Fluid, and Compression Set Resistant (O-rings, Class 1, 75 Hardness).

STANDARDS

DEPARTMENT OF DEFENSE

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment (Type I - Environmental and Type II Internally Induced).
MIL-STD-740 - Airborne and Structureborne Noise Measurements and Acceptance Criteria of Shipboard Equipment.
MIL-STD-1399 - Interface Standard for Shipboard Systems Section 301 Ship Motion and Attitude.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

7.2.2 Non-Government publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents cited in the solicitation (see 7.1).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B 16.5 - Steel Pipe Flanges and Flanged Fittings. (DoD adopted)
ANSI B 93.31 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 167 - Standard Specification for Stainless and Heat-Resisting Chromium-Nickel Steel Plate, Sheet, and Strip. (DoD adopted)
ASTM A 276 - Standard Specification for Stainless and Heat-Resisting Steel Bars and Shapes. (DoD adopted)
ASTM D 3951 - Standard Practice for Commercial Packaging.
ASTM E 437 - Standard Specification for Industrial Wire Cloth and Screens (Square Opening Series).
ASTM F 593 - Standard Specification for Stainless Steel Bolts, Hex Cap Screws, and Studs.
ASTM F 594 - Standard Specification for Stainless Steel Nuts.

(Application for copies should be addressed to the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2859.)

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FLUID CONTROL INSTITUTE (FCI)
78-1 - Pressure Rating Standard for Pipeline Strainers Other than "Y" Type.

(Application for copies should be addressed to the Fluid Control Institute, Inc., P.O. Box 3854, Tequesta, FL 33458.)

INSTRUMENT SOCIETY OF AMERICA (ISA)
S75.02 - Control Valve Capacity Test Procedure.

(Application for copies should be addressed to the Instrument Society of America, 67 Alexander Dr., P.O. Box 12277, Research Triangle Park, NC 27709.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)
J 1926 - Specification for Straight Thread O-ring Boss Port.

(Application for copies should be addressed to the Society of Automotive Engineers Inc., 400 Commonwealth Drive, Warrendale, PA 15096-0001)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through informational service.)

7.2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

7.3 Technical documents. Unless otherwise specified, a technical manual shall be provided. The technical content of these manuals should include sufficient technical details to enable a repair, maintenance, or overhaul activity to restore the equipment dimensions, clearances, and tolerances such that the equipment is capable of perform- ing its function as originally specified and is capable of being logistically supported by the DoD logistics support system.

7.4 Sub-contracted material and parts. The packaging or delivery preparation requirements of referenced documents listed in section 7.2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

7.5 Ordering spare or repair parts. When ordering spare parts or repair parts for the equipment covered by this commercial item description, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

7.5.1 Repair parts and repair parts boxes. Complete sets of repair parts elements for each type and class shall be furnished as follows:

(a) One set elements for each of the ten initial strainers supplied.
(b) One set elements for each two additional strainers supplied.
(c) Cover gaskets - four for every strainer supplied.
(d) Other gaskets, seals and springs - one set for every ten strainers supplied.

When specified (see 7.1) items (c) and (d) shall be furnished in repair boxes.

7.6 Preinstallation instruction. A set of instructions covering the preinstallation of the equipment should be furnished. Instructions should include all information necessary to return the unit to active status, such as, but not limited to: the addition of lubricants prior to operation,

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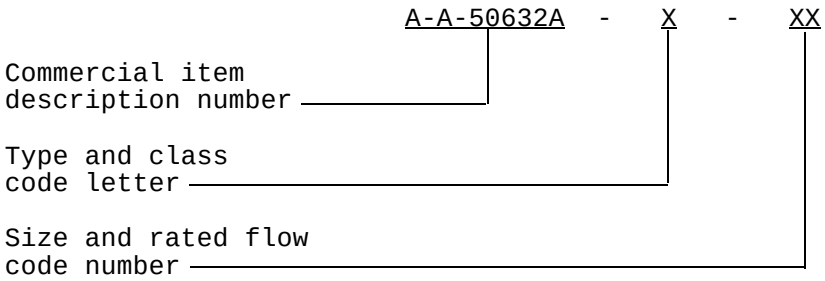
flushing of lines, removal of greaseproof barrier and the location of detached components. Instructions should be packaged in a transparent waterproof plastic bag. Closure should be by heat sealing. The shipping container in which the instructions are packed should be so marked.

7.7 Part number. The part number is a definitive number which corresponds to the type, class, size, and rated flow of strainers covered by this commercial item description. The commercial item description number, the type and class code letter, and the size and rated flow code numbers are combined to form a definitive part number. Part numbers for the strainers are assigned as follows:



7.7.1 Type and class. The type and class of strainer (see 2.1) are identified by a single letter (see table IV).

TABLE IV. Code letter to type and class.

Class	Type		
	I	II	III
1	A	B	C
2	D	E	F
3	G	H	I

7.7.2 Size and rated flow. The size and rated flow of strainer (see 2.1) are identified by a double digit number (see table V).

TABLE V. Code number to size and rated flow.

	Rated flow (gpm)								
Size	75	150	300	450	600	750	900	1050	1200
2	01	-	-	-	-	-	-	-	-
3	-	02	-	-	-	-	-	-	-
4	-	-	03	-	-	-	-	-	-
6	-	-	-	04	05	06	-	-	-
8	-	-	-	-	-	-	07	08	09

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7.8 Subject term (key word) listing.

Continuous
Screen
Cleaning
Filtration
Fittings
Motor
Fuel
Lube oil
Valves

MILITARY INTERESTS:

Custodians:
Army - AT
Navy - SH
Air Force - 16

Preparing activity:
Navy - SH
(Project 4730-0153)

Review activities:
Army - CR, MR
Navy - OS, CG, MC
Air Force - 10
DLA - CS

Civil agency coordinating activities: (where appropriate)
AGR - APS
HHS - FEC
GSA - FSS

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